FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

Hypoglycaemia

Tutorial | Week 4 · Small babies, sugar & the heart | Paediatrics | Dr Mawanda

Hypoglycaemia (blood glucose below 2.5 mmol/L in a well-nourished child, below 3.0 mmol/L in severe malnutrition, below 2.6 mmol/L in Ugandan newborn units) is common in sick Ugandan children with severe malaria, sepsis and malnutrition, kills or brain-damages them within hours, and is cured in minutes with 10% dextrose 5 mL/kg (2 mL/kg in neonates) followed by feeding or a glucose infusion.

Must know before the ward round

1. Treatment thresholds: well-nourished child <2.5 mmol/L (45 mg/dL); SAM <3.0 mmol/L (54 mg/dL); neonate <2.6 mmol/L in Ugandan units (WHO pocket book uses 2.2 mmol/L); child with diabetes <3.9 mmol/L (ISPAD 2022 level 1, 'alert value'). mg/dL ÷ 18 = mmol/L.
2. If you cannot measure glucose in a child with coma or convulsions, treat as hypoglycaemia.
3. 10% dextrose 5 mL/kg IV over 2–5 min for a child; 2 mL/kg for a neonate; recheck glucose after 30 min and repeat if still low (UCG 2023 range: 2–5 mL/kg).
4. Never give 50% dextrose undiluted to a child. 10% = 1 mL/kg of 50% + 4 mL/kg water for injection, or remove 50 mL from a 500 mL bottle of 5% dextrose and add 50 mL of 50% (UCG 2023).
5. No IV access: 10% dextrose or sugar water (20 g sugar in 200 mL) 5 mL/kg by NG tube, or a level teaspoon of moistened sugar under the tongue every 10–20 min; glucagon 0.5 mg (<25 kg) or 1 mg (≥25 kg) IM for insulin-treated children.
6. After the bolus: feed (breast milk, milk, F-75) or run maintenance fluid containing 5–10% dextrose; neonates on day 1: 10% dextrose 60 mL/kg/day.
7. Glucose infusion rate (mg/kg/min) = % dextrose × mL/kg/day ÷ 144. Needing more than 8–10 mg/kg/min suggests hyperinsulinism.
8. Severe malaria: hypoglycaemia is a criterion of severity (UCG: <2.2 mmol/L with parasitaemia), worsened by quinine (stimulates insulin release); check glucose 4-hourly (2-hourly if unconscious or on quinine) and on any deterioration.
9. SAM: every child gets 50 mL of 10% glucose or sugar water on admission, then F-75 every 30 min for 2 h, then every 2 hours day and night; hypoglycaemia and hypothermia travel together (and signal sepsis).
10. Commonest cause of recurrent hypoglycaemia in a well toddler (18 months–5 years) is ketotic hypoglycaemia after an overnight fast or illness; low glucose WITHOUT ketones means hyperinsulinism or a fatty-acid oxidation defect.
11. Take a 'critical sample' (glucose, ketones, insulin, cortisol, growth hormone, lactate) before dextrose if the cause is unclear, but never delay treatment for it.

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1. Definition

Hypoglycaemia means a blood glucose level low enough to harm the brain. Glucose is the brain's main fuel, and a child's brain is large compared with the body, so children use **two to three times more glucose per kg** than adults and have **small glycogen stores**. A sick, fasting or malnourished child can run out within hours.

It is **easy to miss** (it looks like the illness itself: drowsiness, convulsions, coma), **easy to diagnose** (a glucometer reading in 30 seconds) and **easy to treat** (10% dextrose). Missing it causes **brain damage, epilepsy and death**.

There is no single "normal" cut-off. We use **operational thresholds**: the level at which we act.

Group	Treat when glucose is below	Source / comment
Well-nourished child (ETAT, sick child)	**2.5 mmol/L (45 mg/dL)**	WHO Pocket Book 2013, ETAT
Child with SAM	**3.0 mmol/L (54 mg/dL)**	WHO SAM protocol, UCG 2023 (chapter 19)
Severe malaria (criterion of severity)	**2.2 mmol/L (40 mg/dL)**	UCG 2023 malaria table; WHO malaria guidelines
General (UCG 2023 section 1.1.6)	Symptoms usually occur **<3.0 mmol/L** (55 mg/dL)	Adults and children
Neonate	**2.6 mmol/L (47 mg/dL)** in Ugandan newborn units; WHO Pocket Book treats **<2.2 mmol/L**	See the [!UGANDA] box in section 2 and the Neonatal Resuscitation topic
Child with diabetes (ISPAD 2022)	**Level 1 <3.9 mmol/L** (alert, treat with fast sugar); **Level 2 <3.0 mmol/L** (clinically significant); **Level 3** = severe event needing another person's help (coma/convulsion)	ISPAD Clinical Practice Consensus Guidelines 2022
Persistent/recurrent hypoglycaemia work-up	Keep glucose **>3.3 mmol/L (60 mg/dL)** in the first 48 h of life, and **>3.9 mmol/L (70 mg/dL)** after 48 h if a persistent disorder is suspected	Pediatric Endocrine Society 2015 (specialist targets)

Conversion: mmol/L × 18 = mg/dL (2.5 mmol/L = 45 mg/dL; 3.0 = 54; 2.2 = 40); mg/dL ÷ 18 = mmol/L.

Whipple's triad (used to confirm a true hypoglycaemic disorder in older children): (1) symptoms of hypoglycaemia, (2) a low plasma glucose measured at that time, and (3) relief of symptoms when glucose is raised.

> CLINICAL PEARL
> The one thing never to forget: **check the glucose in every child who is unconscious, convulsing, severely malnourished, in shock or unable to feed. If you cannot check it, treat it**: 10% dextrose 5 mL/kg IV (2 mL/kg in a newborn).

2. Epidemiology

- In Uganda, hypoglycaemia is mostly seen in children with **severe malaria, sepsis, severe acute malnutrition (SAM)**, in **newborns** (preterm, small, cold, asphyxiated, babies of diabetic mothers), and in

children who have **not eaten** for many hours. It is also a feature of **poisoning** (alcohol, sulphonylureas), **insulin treatment** in type 1 diabetes, and rare endocrine and metabolic diseases.

- **Severe malaria**: hypoglycaemia is found in roughly **one in ten or more** children admitted with severe malaria in African studies, and many more in those in coma; it is associated with a several-fold increase in mortality. Quinine used to make this worse; IV **artesunate** (now first line in Uganda) causes much less hypoglycaemia.
- **SAM**: most deaths in the first days of admission come from the trio of **hypoglycaemia, hypothermia and infection**. That is why the WHO/UCG protocol feeds every 2–3 hours, day and night.
- **Newborns**: preterm/LBW babies, babies of mothers with diabetes or GDM (rising in Uganda), asphyxiated, cold or septic babies, and babies who are not breastfed early.
- **Children with type 1 diabetes**: hypoglycaemia is the commonest acute complication of insulin therapy; families often lack glucometers, strips or reliable food supply.
- **Critically ill children**: Ugandan studies (e.g. at Fort Portal Regional Referral Hospital) found **dysglycaemia (low or high glucose) in a substantial proportion of emergency admissions, linked to higher mortality**.

> UGANDA CONTEXT
> Practical realities: **glucometers** are available at most HC IVs and hospitals but **strips often run out**; 10% dextrose is usually **made on the ward** from 50% dextrose and water for injection or 5% dextrose; NG tubes and sugar are always available. Neonatal units in Uganda (Mulago, Mengo, RRHs) generally treat newborns at **<2.6 mmol/L**, which is what the Neonatal Resuscitation topic uses; the WHO Pocket Book uses 2.2 mmol/L. When in doubt in a symptomatic baby, treat.

3. Aetiology and Risk Factors

Classification by cause

Mechanism	Examples
Too little supply	Fasting, vomiting, poor feeding, SAM, delayed breastfeeding in newborns, malabsorption
Too much use	Sepsis, **malaria** (parasites consume glucose), fever, hypothermia, convulsions, respiratory distress, asphyxia
Too little production (liver)	Prematurity/SGA (low glycogen), liver failure, glycogen storage disease, galactosaemia, hereditary fructose intolerance, gluconeogenesis defects, **alcohol** (blocks gluconeogenesis)
Hormone deficiency	**Cortisol** (CAH, Addison's, sudden stop of steroids, hypopituitarism), **growth hormone** (hypopituitarism: think micropenis + prolonged jaundice + hypoglycaemia in a newborn)
Too much insulin (hyperinsulinism)	**Infant of a diabetic mother**, **quinine**, insulin overdose in diabetes, **sulphonylurea ingestion** (grandparent's glibenclamide), congenital hyperinsulinism, Beckwith–Wiedemann syndrome, insulinoma (rare), perinatal stress hyperinsulinism (SGA, asphyxia)
Ketotic hypoglycaemia	Thin toddlers (18 months–5 years) who become hypoglycaemic after an overnight fast or during an intercurrent illness; ketones high, insulin suppressed; a diagnosis of exclusion. The commonest cause of recurrent hypoglycaemia in a well toddler
Fatty-acid oxidation defects	Hypoglycaemia **without ketones** during fasting/illness (e.g. MCAD deficiency); may present as "sudden infant death"

Who to check (high-risk groups): every child who is unconscious, convulsing, in shock, unable to feed or severely malnourished; severe malaria (especially on quinine); sepsis; newborns at risk (preterm, LBW/SGA, infant of a diabetic mother, asphyxia, sepsis, hypothermia, poor feeding); children with diabetes on insulin;

suspected ingestion (alcohol, diabetes tablets); children kept "nil by mouth".

4. Pathophysiology

How the body keeps glucose up

Phase	Time after a meal	Main glucose source
Fed	0–4 h	Glucose from the gut; insulin high → glycogen stored
Post-absorptive	4–12 h	**Glycogenolysis** (liver glycogen): lasts only ~8–12 h in an infant, much less in a preterm or SGA baby
Fasting	>12 h	**Gluconeogenesis** (from lactate, alanine, glycerol) + **lipolysis** → fatty acids → **ketones** (brain fuel)

Counter-regulatory hormones: **glucagon, adrenaline, cortisol, growth hormone**. Insulin is the only hormone that **lowers** glucose. Normally, as glucose falls, insulin switches **off**, fat breakdown switches **on** and ketones rise. So:

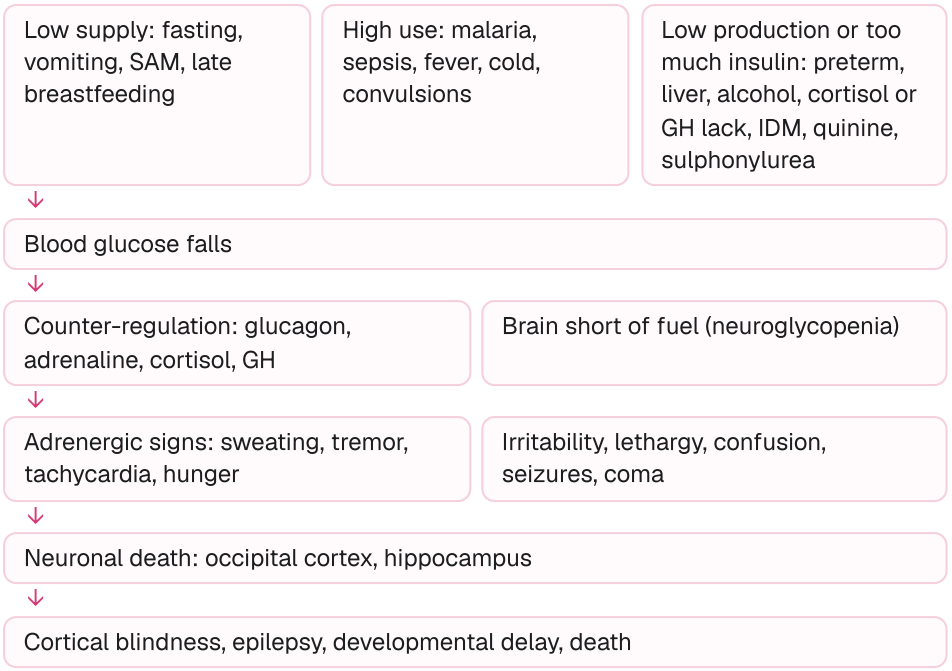
- **Low glucose + high ketones** = the body is responding normally to starvation (ketotic hypoglycaemia, SAM, prolonged fasting) or there is a hormone deficiency (cortisol/GH) or glycogen storage disease.
- **Low glucose + no ketones** = insulin is still switched on (**hyperinsulinism**: IDM, quinine, sulphonylurea, insulin) or the child cannot burn fat (**fatty-acid oxidation defect**).

Why malaria causes hypoglycaemia

1. **Reduced intake**: vomiting, anorexia, fasting.
2. **Increased use**: fever, parasites (which metabolise glucose anaerobically), and hypoxic tissues.
3. **Impaired gluconeogenesis**: the liver is overwhelmed; lactic acidosis.
4. **Quinine** stimulates the pancreatic beta cells to **release insulin** (hyperinsulinaemic hypoglycaemia), especially in pregnancy and in children.

Why newborns are vulnerable: at birth the continuous glucose supply through the placenta stops. Glucose normally **falls in the first 1–2 hours** and then rises as feeding and glycogenolysis begin ("transitional hypoglycaemia"). Babies with **small glycogen stores** (preterm, SGA), **high use** (cold, septic, asphyxiated, respiratory distress) or **high insulin** (infant of a diabetic mother, because the fetal pancreas has been making extra insulin) cannot cope.

Why the brain is damaged: neurons cannot store glucose. Severe or prolonged hypoglycaemia causes **neuronal death**, particularly in the **occipital cortex** (newborns: cortical blindness, epilepsy) and the hippocampus. Recurrent neonatal hypoglycaemia is linked to later developmental delay and learning problems.



Feature links

- Falling glucose → adrenaline release → **sweating, tremor, pallor, tachycardia, anxiety**; hunger signals → **hunger**.
- Brain short of fuel → **irritability, lethargy, confusion, abnormal behaviour, headache, slurred speech, seizures, coma, focal deficits**.
- Newborn brain short of fuel with immature responses → **jitteriness, poor suck, floppiness, apnoea, cyanotic spells, weak or high-pitched cry, seizures**, or **nothing at all**.
- No glycogen and blunted adrenergic response (SAM, beta-blockers, recurrent episodes) → **no warning signs**; sudden coma.
- High metabolic use with failing heat production → **hypoglycaemia travels with hypothermia** (and signals sepsis).
- Fetal hyperinsulinism (IDM) → **macrosomia** and hypoglycaemia in the first 24–72 h.
- Cortisol lack (CAH, adrenal crisis) → **hypoglycaemia with shock, low Na, high K**, ambiguous genitalia, hyperpigmentation.
- GH and ACTH lack (hypopituitarism) → **micropenis, midline defects, prolonged jaundice** with hypoglycaemia.
- Glycogen storage disease → **hepatomegaly**, high lactate.
- Hyperinsulinism → **no ketones** and a **high glucose infusion rate** (>8–10 mg/kg/min) needed.
- Occipital neuronal death → **cortical visual impairment**, later epilepsy.

5. Classification / Staging

- **By severity in diabetes (ISPAD 2022)**: **Level 1** <3.9 mmol/L (alert value); **Level 2** <3.0 mmol/L (clinically significant); **Level 3** severe event needing another person's help (coma or convulsion).
- **By mechanism**: supply, use, production, hormone deficiency, hyperinsulinism, ketotic, fatty-acid oxidation (table in section 3).
- **By ketones**: **ketotic** (starvation, SAM, ketotic hypoglycaemia, cortisol/GH deficiency, glycogen storage disease) versus **non-ketotic** (hyperinsulinism, fatty-acid oxidation defect).
- **Neonatal**: **transitional** (first 1–2 hours, resolves with feeding) versus **persistent** (beyond 48–72 h, or needing GIR >8–10 mg/kg/min: hyperinsulinism, hypopituitarism, adrenal insufficiency, inborn errors).

- **Symptoms by mechanism**:

Adrenergic (early; glucose falling)	Neuroglycopenic (brain short of fuel)
Sweating, tremor, pallor, tachycardia, anxiety, hunger	Irritability, lethargy, confusion, abnormal behaviour, headache, slurred speech, seizures, coma, focal deficits

6. Clinical Features

History

Question	Why it matters
Age, weight, gestation and birth weight (neonate)	Preterm/SGA babies have little glycogen; doses per kg
When did the child last eat or breastfeed? Vomiting? How long fasting?	Fasting is the commonest trigger; fasting-related (ketotic, SAM) versus inappropriate (hyperinsulinism if hypoglycaemic soon after feeding)
Fever, convulsions, reduced consciousness; malaria test result; antimalarial given (quinine?)	Malaria, sepsis, meningitis; quinine → hyperinsulinism
Medicines: quinine, insulin, any adult diabetes medicines in the home (glibenclamide, gliclazide, metformin), beta-blockers, salicylates, herbal medicines	Drug-induced hyperinsulinism or toxicity; sulphonylurea needs prolonged observation (24 h)
Alcohol (local brew, waragi, sachets)	Alcohol blocks gluconeogenesis; small children become hypoglycaemic and hypothermic
Mother has diabetes or GDM? Big baby? Asphyxia, temperature, feeding (newborn)	Infant of a diabetic mother → hyperinsulinism in the first 24–72 h; transitional versus persistent
Previous episodes; relation to fasting or meals; time since last meal at each episode	Recurrent fasting hypoglycaemia → ketotic, endocrine or metabolic cause; needs investigation
Growth, weight loss, diarrhoea, oedema	SAM
Known diabetes: insulin dose, last injection, meal skipped, exercise	Diabetes-related hypoglycaemia; plan for education
Family history: unexplained infant deaths, consanguinity	Inborn errors of metabolism
Prolonged jaundice, micropenis, midline defects (newborn)	Hypopituitarism
HIV status, immunisation, feeding	Background risk and sepsis

Examination

Follow the order of Hutchison's Clinical Methods (24th edition), adapted for children. In practice the bedside glucose comes in the first minute of the ABCDE assessment (section 9); complete the full examination once glucose is corrected.

Introduction, consent and preparation

- Introduce yourself; confirm the child's name, age and the carer's relationship.
- Explain; consent from the parent and assent from the child (where possible).
- **Hand hygiene**; position (recovery position if unconscious); keep warm, especially newborns and malnourished children (expose only what you need).

General inspection (end of the bed)

- **Conscious level** (AVPU or Blantyre coma score) before and after dextrose; convulsions.
- Sweating, tremor, pallor (adrenergic) versus lethargy, abnormal behaviour (neuroglycopenic).

- **Nutrition**: wasting, **oedema** (SAM); weight, MUAC (6–59 months), Z-scores.
- **Hydration**, jaundice, pallor.
- **Vital signs**: **temperature (hypothermia)**, HR, RR, SpO_2, CRT; **bedside glucose** with the time and method recorded.
- **Smell of alcohol**.
- **Newborns**: jitteriness, poor suck, floppiness, apnoea, cyanotic spells, hypothermia, high-pitched or weak cry, seizures, or **nothing at all** (screening at-risk babies is therefore essential); macrosomia (IDM).

Hands, face and neck

- Sweating, tremor, cold peripheries.
- **Hyperpigmentation** of creases and gums (adrenal insufficiency).
- **Midline facial defects** (hypopituitarism); **macroglossia** (Beckwith–Wiedemann).
- Neck stiffness, bulging fontanelle (meningitis as an alternative or coexisting cause).

The system: abdomen, genitalia and neurology

- **Abdomen**: **hepatomegaly** (glycogen storage disease, heart failure, liver failure), **splenomegaly** (malaria), omphalocele (Beckwith–Wiedemann).
- **Genitalia**: **ambiguous genitalia** (CAH), **micropenis** (hypopituitarism).
- **Skin**: **insulin injection sites** (lipohypertrophy), signs of sepsis.
- **Neurology**: focal deficits, tone, pupils.

Completing the examination

- Recheck glucose 30 minutes after treatment.
- Thank, re-dress, **hand hygiene**, summarise.

Document these positives	And these negatives
Glucose value, time, and the method (glucometer versus lab)	"No hepatomegaly"
AVPU/Blantyre score before and after dextrose	"Genitalia normal; no hyperpigmentation"
Temperature, weight, MUAC, oedema	"No signs of fluid overload"
Time and volume of each dextrose bolus	"Glucose rechecked at 30 min: x mmol/L"

Red flags

RED FLAG

- Coma or convulsions **not improving within 15–30 minutes** of correcting the glucose: look for another cause (cerebral malaria, meningitis, poisoning, head injury).
- Hypoglycaemia that **recurs** despite feeds and a 10% dextrose infusion: needs a higher glucose infusion rate and senior input (hyperinsulinism, sepsis, adrenal insufficiency).
- **Hypoglycaemia + hypothermia** in a malnourished or newborn child = sepsis until proven otherwise.
- Hypoglycaemia with **shock, hyponatraemia and hyperkalaemia** in an infant (± ambiguous genitalia or hyperpigmentation) = **adrenal crisis**: give hydrocortisone.
- Hypoglycaemia in a child **without** ketones: hyperinsulinism or a fatty-acid oxidation defect.

Thinking like a clinician

- **"Treat first, think second."** The bolus takes 2 minutes; the work-up can wait.
- **"Why is this child hypoglycaemic?"** is the second question. In Uganda the answer is usually **malaria, sepsis, SAM, fasting, or the newborn period**. If none fits, or it keeps recurring, think of drugs

(sulphonylurea, alcohol), hormones (cortisol, GH), the liver, and hyperinsulinism.

- **Unconscious child with malaria whose glucose is now 6 mmol/L but who is still in coma**: the hypoglycaemia was not the whole story (cerebral malaria, post-ictal state, meningitis). Keep checking glucose 2–4 hourly.
- **A baby who needs a lot of glucose** (glucose infusion rate >8–10 mg/kg/min) to stay normal almost certainly has **hyperinsulinism**.
- **Glucagon works only when the liver has glycogen**: useful for insulin overdose in a well-fed diabetic child; useless in SAM, prolonged fasting, alcohol, or liver disease.
- **Common student mistakes**: not checking glucose in a convulsing child; giving 50% dextrose undiluted (vein damage, rebound); giving a bolus but not following it with feeds or a dextrose infusion (rebound hypoglycaemia); forgetting to recheck at 30 minutes; using the adult threshold of 4 mmol/L to decide on a bolus in a non-diabetic sick child (fine to be cautious, but know the numbers); using glucagon in SAM.

EXAM FAVOURITE

Viva favourite: **"Why does quinine cause hypoglycaemia, and how do you prevent it?"** Quinine stimulates insulin secretion from pancreatic beta cells. Give quinine diluted in **5% or 10% dextrose**, feed the child, check glucose every 2–4 hours, and prefer **IV artesunate** (UCG 2023 first line for severe malaria).

7. Differential Diagnoses

When a child presents with coma, convulsions or abnormal behaviour, hypoglycaemia is one cause among several, and often coexists with them.

Differential	Distinguishing clinical feature	Distinguishing investigation
Hypoglycaemia	Rapid improvement after dextrose	Glucose
Cerebral malaria	Fever; coma persists after glucose corrected	Positive mRDT/smear; glucose
Bacterial meningitis	Neck stiffness, bulging fontanelle, purpura	LP (when safe)
Febrile convulsion	Age 6 months–5 years, brief generalised seizure, full recovery	Glucose normal; exclude meningitis/malaria
Post-ictal state	Recovers within about 1 hour	Observation
Poisoning (alcohol, sulphonylurea, organophosphate, sedatives)	History, smell, pupils, secretions	History; glucose (alcohol and sulphonylureas cause hypoglycaemia)
DKA / hyperglycaemia	Polyuria, acidotic breathing	High glucose, urine ketones
Hypocalcaemia (neonate, rickets)	Tetany, jitteriness, seizures	Calcium low, glucose normal
Hyponatraemia	Diarrhoea, hypotonic fluids, seizures	Sodium
Neonatal jitteriness (benign)	Stops with gentle restraint, baby feeds well	Glucose normal
Adrenal crisis	Shock, ± ambiguous genitalia/pigmentation	Low Na, high K; cortisol

8. Clinical Workup

"Available at" is the lowest level that usually offers the test Verify *availability compiled from the source notes, not an MoH list*.

Bedside

Test	Why	Expected finding	Available at
Glucometer glucose (heel prick in infants, finger prick in older children)	Fast diagnosis	Below threshold; less accurate at very low levels (confirm with a lab glucose if possible, but **never wait for it** before treating)	Most HC IVs and hospitals (strips often run out)
Urine or blood ketones (dipstick, BOHB meter)	Mechanism	High (starvation, ketotic hypoglycaemia, hormone deficiency) versus absent (hyperinsulinism, fatty-acid oxidation defect)	Dipstick at HC IV; BOHB meters rare
mRDT / malaria smear	Commonest cause in febrile Ugandan children	Parasitaemia → IV artesunate; glucose 2–4-hourly	All levels
Hb (HemoCue), **temperature**	Anaemia; hypothermia (sepsis)	Low Hb; low temperature	All levels (HemoCue variable)
MUAC (6–59 months)	SAM	<11.5 cm = SAM	All levels

Basic lab

Test	When	Expected / abnormal finding	How it changes management	Available at
Laboratory plasma glucose	To confirm a glucometer reading	<2.5 mmol/L (or local threshold)	Confirms the diagnosis	Hospital labs
Blood culture, FBC, CRP	Fever, hypothermia, SAM, newborn	Sepsis (hypoglycaemia + hypothermia = sepsis)	Antibiotics	RRH/national referral (FBC at hospitals)
Electrolytes (Na, K)	Suspected adrenal crisis, vomiting, dehydration	Low Na, high K	Hydrocortisone, saline	RRH/national referral
LFTs, INR	Hepatomegaly, jaundice	Liver failure	Supportive, referral	Hospitals
HIV test	Status unknown	Positive	Background risk for sepsis, SAM	All levels

Imaging

Not applicable for the diagnosis of hypoglycaemia itself; imaging is used only for the cause or alternative diagnoses (e.g. cranial imaging for a child still in coma after glucose correction, abdominal ultrasound for hepatomegaly).

Specialised / referral-level

Test	When	Expected / abnormal finding	How it changes management	Available at
Critical sample: insulin, C-peptide, cortisol, growth hormone, BOHB, free fatty acids, lactate, ammonia; first urine for ketones and organic acids	Unexplained or recurrent hypoglycaemia (no malaria, no SAM, not a newborn, not diabetic), taken **at the time of hypoglycaemia, with the cannula, before the dextrose**, but never delaying treatment	Insulin detectable when glucose low = hyperinsulinism; low cortisol = adrenal insufficiency; high lactate = gluconeogenesis defect/GSD	Specific treatment (diazoxide, hydrocortisone, diet)	Mulago/private labs or send-out; often unavailable. Freeze serum if possible

Test	When	Expected / abnormal finding	How it changes management	Available at
Sulphonylurea screen, alcohol level	Suspected ingestion	Positive	Observe ≥24 h; octreotide	Rarely available

Putting it together

Model one-line summary: "A 3-year-old girl, 12 kg, with severe malaria and **hypoglycaemia (1.8 mmol/L)** causing a convulsion; corrected with 10% dextrose 60 mL, now 5.6 mmol/L on maintenance 10% dextrose-containing fluid and IV artesunate."

Problem list

1. Hypoglycaemia.
2. Its cause (severe malaria).
3. Risk of recurrence.
4. Other complications (anaemia, acidosis, seizures).

Model case presentation (ward round)

> "Baby A.K., a **2-year-6-month-old girl** from Kawempe, **weight 11.5 kg** (weight-for-height Z-score −1, height-for-age Z −1.5, MUAC 13.5 cm, no oedema), admitted at 03:00 with **fever for 2 days, vomiting and a generalised convulsion at home**. On arrival she was **AVPU P**, glucose **1.6 mmol/L**, mRDT positive (smear 3+), Hb 8.2 g/dL. She received **10% dextrose 57 mL IV** (5 mL/kg) and rectal diazepam 5 mg; glucose at 30 minutes was **5.9 mmol/L** and she woke to voice within an hour. She is on **IV artesunate 34 mg (3 mg/kg)** and **maintenance 10% dextrose in 0.45% saline at 43 mL/h**. Glucose has been 4.8–6.2 mmol/L on 4-hourly checks. She is **fully immunised for age** (child health card), **eats family food, breastfed until 18 months**; **mother HIV-negative** in this pregnancy, child's HIV test today negative. **Assessment: severe malaria with hypoglycaemia and a convulsion, now stable**. **Plan**: continue artesunate to 3 doses then oral ACT, start oral feeds and reduce IV fluid, 4-hourly glucose until feeding well for 24 h, Hb tomorrow, counsel on bed nets and early care-seeking."

NOTE

Worked example 1: calculating a glucose infusion rate (GIR). A 2.0 kg neonate is on 10% dextrose at 90 mL/kg/day and is still hypoglycaemic. **GIR = % dextrose × mL/kg/day ÷ 144 = 10 × 90 ÷ 144 = 6.25 mg/kg/min.** Normal neonatal glucose production is about 4–6 mg/kg/min. Increase the fluid volume (if the baby tolerates it) or the concentration (12.5% through a peripheral line at most; higher concentrations need a central line). If the baby needs **>8–10 mg/kg/min**, suspect **hyperinsulinism** and discuss with a paediatrician.

NOTE

Worked example 2: ketones tell you the mechanism. A 2-year-old has a seizure at 7 a.m. after missing supper because of a cold; glucose 2.1 mmol/L, **urine ketones +++**, thin, no hepatomegaly. → **Ketotic hypoglycaemia** (diagnosis of exclusion). Plan: dextrose, then advise frequent carbohydrate snacks, a bedtime snack, and sugary drinks during illness; check ketones at home if possible. If **ketones were negative**, you would worry about hyperinsulinism or a fatty-acid oxidation defect and refer.

9. Management

Immediate / emergency

First 5 minutes (ABCDE with glucose in D)

1. **A and B** (ETAT): open the airway, recovery position if unconscious, oxygen if SpO_2 <94% with emergency signs.
2. **C**: circulation, IV or IO access.
3. **D: check glucose** (glucometer, heel prick in infants, finger prick in older children).
4. **If <2.5 mmol/L (or <3.0 in SAM, <2.6 in a neonate), or you cannot measure it**:
   - IV access → **10% dextrose 5 mL/kg** (neonate **2 mL/kg**) over 2–5 minutes.
   - No IV → NG 10% dextrose/sugar water 5 mL/kg, sublingual sugar, or IO; glucagon if insulin-induced.
5. **If convulsing**, give the anticonvulsant as well (rectal diazepam 0.5 mg/kg; phenobarbital in neonates).
6. **If the cause is unclear** (no malaria, no SAM, not a newborn, not diabetic), draw a **critical sample with the cannula, before the dextrose**, but never let it delay treatment.
7. **E**: temperature; keep warm.
8. **Recheck glucose after 30 minutes** (15 minutes in a diabetic child). If still low, repeat the bolus.

> **RED FLAG**
> Call your senior if: glucose stays low after two boluses; the child remains unconscious after glucose is normal; there is shock or suspected adrenal crisis; a newborn needs repeated boluses.

Emergency treatment algorithm

```
Sick child: coma / convulsing / SAM / shock / not feeding / newborn at risk
                          |
                CHECK GLUCOSE (or cannot check)
                          |
      Low* or unknown? ──┴── Normal → look for other causes; recheck 4-hourly if at risk
            |
   Conscious & can swallow? ── yes ──► oral sugar/feed (0.3 g/kg; breastfeed; milk; F-75)
            | no
   IV/IO access? ── yes ──► 10% DEXTROSE 5 mL/kg (neonate 2 mL/kg) over 2–5 min
            | no
   NG tube: 10% dextrose or sugar water 5 mL/kg, or milk
   + sublingual sugar every 10–20 min; glucagon if insulin-treated
            |
   RECHECK at 30 min ── still low? → repeat bolus; senior review
            |
   Prevent rebound: feed or start dextrose infusion; treat the cause
   Monitor 1–4 hourly until stable

*<2.5 mmol/L child · <3.0 mmol/L SAM · <2.6 mmol/L neonate (Uganda) · <3.9 mmol/L diabetic
```

Definitive

1. **Prevent rebound**: feed (breast milk, milk, F-75) or run maintenance fluid containing 5–10% dextrose straight after the bolus.
2. **Treat the cause**: antimalarial, antibiotics, SAM protocol, insulin adjustment, hydrocortisone for adrenal crisis.
3. **Monitor** glucose 1–4-hourly until stable and feeding.
4. **Investigate** recurrent or unexplained hypoglycaemia (critical sample, ketones) and refer.

Hypoglycaemia in severe malaria

1. Check glucose **on admission** in every child with severe malaria, then **every 4 hours** (every 2 hours if unconscious, convulsing or on quinine), and at any deterioration.
2. Treat <2.5 mmol/L (UCG severity criterion <2.2 mmol/L) with **10% dextrose 5 mL/kg over 5–7 minutes**; recheck at 30 min.
3. Give **maintenance fluid containing dextrose** (e.g. 5% or 10% dextrose in 0.45% saline) if not feeding; start feeds (breastfeeding or NG milk) as soon as safe.
4. Give **IV/IM artesunate**: 3 mg/kg if <20 kg, 2.4 mg/kg if ≥20 kg, at 0, 12 and 24 h, then daily until able to take oral ACT.
5. If quinine must be used (artesunate unavailable): **10 mg/kg in 5–10% dextrose** over 4 hours, 8-hourly (UCG gives quinine in dextrose 5%), with 2-hourly glucose checks.

Hypoglycaemia in severe acute malnutrition

1. **All** SAM children are treated as at risk: on admission give **50 mL of 10% glucose or sugar water** (1 rounded teaspoon of sugar in 3 tablespoons of water, UCG 2023) orally or by NG, followed by the first feed as soon as possible.
2. If able to drink: **F-75 every 30 minutes for 2 hours**, then **every 2 hours for 24 hours** (day and night), then every 2–3 hours.
3. If unconscious or convulsing: **10% glucose 5 mL/kg IV**, then 50 mL of 10% glucose or sucrose by NG, then F-75 by NG.
4. Recheck glucose after 30 minutes and every 2–4 hours on day 1; if it falls below 3 mmol/L again, repeat the 50 mL glucose and continue 2-hourly feeds.
5. Keep warm (skin-to-skin, cover the head), and start **antibiotics** (hypoglycaemia in SAM usually means infection).
6. **Do not use glucagon** (no glycogen).

Neonatal hypoglycaemia

1. **Prevent**: breastfeed within 1 hour of birth, then every 2–3 hours (at least 8 times/24 h); keep warm (skin-to-skin, KMC).
2. **Screen at-risk babies** (preterm, LBW/SGA <2.5 kg, infant of a diabetic mother, asphyxia, sepsis, hypothermia, poor feeding): glucose before the second feed (2–4 h of age), then before feeds for at least 24 hours (IDM, SGA) or until stable.
3. **Asymptomatic, 2.0–2.6 mmol/L, able to feed**: breastfeed or give expressed breast milk (cup or NG, about 10 mL/kg), recheck in 30–60 min. If still low → IV dextrose.
4. **Symptomatic, or <2.0–2.2 mmol/L, or not rising with feeds**: **10% dextrose 2 mL/kg IV**, then **10% dextrose infusion 60 mL/kg/day** on day 1 (higher on later days: about 80, 100, 120, 150 mL/kg/day from day 2 to day 5, per unit protocol); continue enteral feeds if tolerated; recheck 30 min after the bolus, then 1–3 hourly until stable.
5. Persistent need for GIR **>8–10 mg/kg/min** or hypoglycaemia beyond 48–72 h → think hyperinsulinism, hypopituitarism, adrenal insufficiency, inborn errors; refer.

Hypoglycaemia in a child with type 1 diabetes (ISPAD 2022)

1. **Conscious**: give **0.3 g/kg** fast-acting glucose (about 10–15 g: 3 glucose tablets, 150 mL juice or soda, 2–3 teaspoons sugar in water). Recheck in **15 minutes**; repeat if still <3.9 mmol/L ("rule of 15").
2. Then a **snack with complex carbohydrate** (bread, posho, banana) if the next meal is more than an hour away.
3. **Unconscious or convulsing**: **glucagon 0.5 mg (<25 kg) or 1 mg (≥25 kg) IM/SC** at home, or **10% dextrose 2–5 mL/kg IV** in hospital (NICE: maximum 5 mL/kg), then a dextrose infusion until able to eat.
4. **Find the cause**: missed meal, extra exercise, insulin error, alcohol, new illness (vomiting). Adjust insulin, educate the family, and make sure there is sugar at home and at school.

Drug table

Drug	Dose (paediatric per kg, max; adult)	Route	Frequency	Duration	Key cautions
10% dextrose (child)	**5 mL/kg** (UCG 2023 range 2–5 mL/kg) = 0.5 g/kg. WHO weight bands: <4 kg 15 mL; 4–6 kg 25 mL; 6–10 kg 40 mL; 10–14 kg 60 mL; 14–19 kg 80 mL	IV/IO over 2–5 min (UCG malaria table: 5–7 min)	Recheck at 30 min; repeat once if still low	Bolus, then infusion or feed	**Never give 50% undiluted to a child** (sclerosis, tissue necrosis, rebound)
10% dextrose (neonate)	**2 mL/kg** (0.2 g/kg)	IV over 5 min	Recheck at 30 min	Bolus	Follow with infusion 60 mL/kg/day (day 1), increasing daily
Making 10% dextrose	1 mL/kg of 50% + 4 mL/kg water for injection; or remove 50 mL from 500 mL of 5% dextrose and add 50 mL of 50% (UCG 2023)	n/a	Make fresh	Do not store	Write the volume of 10% on the chart
Oral glucose / sugar (conscious child)	**0.3 g/kg** fast-acting sugar (about 10–20 g: 2–4 teaspoons sugar in 100–200 mL water; UCG 2023)	Oral	Repeat after 15 min if still low	Then a snack or meal	Only if able to swallow
Sugar water via NG	5 mL/kg of sugar solution (4 level teaspoons = 20 g sugar in 200 mL clean water) or milk	NG tube	Then regular feeds	Until IV access or feeding	When IV access fails
Sublingual sugar	1 level teaspoon of sugar moistened with a few drops of water	Under the tongue	Every 10–20 min	Until IV or feeds	First aid when IV is impossible or delayed (WHO Pocket Book)
Glucagon	**0.5 mg if <25 kg (or <6–8 years); 1 mg if ≥25 kg** (about 20–30 µg/kg); adult 1 mg	IM or SC	Once; may repeat after 15 min (UCG: once or twice)	Single dose	For insulin-induced hypoglycaemia when oral/IV not possible. **Not effective in SAM, starvation, alcohol, liver disease**. Causes vomiting
Maintenance fluid with dextrose	Holliday–Segar volume (100 mL/kg/day for the first 10 kg, 50 for the next 10, 20 above 20 kg) of 5–10% dextrose with 0.45%/0.9% saline	IV	Continuous	Until feeding	Prevents rebound; watch for overload
Neonatal 10% dextrose infusion	**60 mL/kg/day** day 1, then about 80, 100, 120, 150 mL/kg/day (days 2–5)	IV	Continuous	Until feeding and stable	Max 12.5% peripherally; GIR >8–10 mg/kg/min = hyperinsulinism
Hydrocortisone (suspected adrenal insufficiency)	2–4 mg/kg (or 25 mg infant; 50 mg 1–5 years; 100 mg >5 years)	IV/IM	Then 6-hourly	Until stable	Shock + low Na + high K ± ambiguous genitalia/pigmentation
Diazoxide (hyperinsulinism, specialist)	5–15 mg/kg/day	Oral	Divided 8–12-hourly	Long term	Specialist only; with a thiazide to limit fluid retention

Drug	Dose (paediatric per kg, max; adult)	Route	Frequency	Duration	Key cautions
Octreotide (sulphonylurea poisoning, specialist)	1–2 µg/kg (max 50 µg)	SC	6–12-hourly	≥24 h	Prevents recurrent hypoglycaemia from sulphonylureas
Artesunate (severe malaria)	**3 mg/kg if <20 kg; 2.4 mg/kg if ≥20 kg**	IV/IM	0, 12, 24 h, then daily	Until oral ACT	Much less hypoglycaemia than quinine
Quinine (only if artesunate unavailable)	**10 mg/kg** in 5–10% dextrose	IV over 4 h	8-hourly	Until oral	Stimulates insulin release: 2-hourly glucose
Diazepam (convulsions)	**0.5 mg/kg**	Rectal	Once, repeat once	Acute	Phenobarbital in neonates

DRUG DOSING

5 mL/kg of 10% dextrose = 0.5 g/kg glucose. In a 12 kg child: 60 mL of 10% = 12 mL of 50% + 48 mL water for injection. Write the volume of 10% on the chart, not "50% dextrose 1 ampoule".

Referral criteria

Level	What to do	Refer up when
Home / VHT	Sugar water or feeds; sublingual sugar; glucagon if a diabetic child	Drowsy, convulsing or not feeding → nearest facility
HC III	Glucometer (if strips), 10% dextrose (made from 50%), NG sugar water, sublingual sugar; treat malaria; SAM first feed	Recurrent hypoglycaemia, coma not resolving, SAM with complications, sick newborn → HC IV or hospital
HC IV / General Hospital	IV dextrose infusions, IV artesunate, SAM inpatient care, neonatal unit	Hypoglycaemia recurring despite infusion, suspected adrenal crisis, persistent neonatal hypoglycaemia, poisoning needing octreotide → RRH
RRH	Electrolytes, blood culture, higher GIR, hydrocortisone, paediatrician	Unexplained or recurrent hypoglycaemia needing a critical sample and endocrine work-up; hyperinsulinism → NRH
NRH (Mulago)	Critical sample analysis, diazoxide, specialist endocrine and metabolic care	n/a

Level split adapted from UCG 2023 levels of care Verify *UCG level codes*.

Monitoring and follow-up

- **Recheck glucose at 30 minutes** after every bolus (15 minutes in diabetes), then **1–4-hourly** until stable and feeding.
- **Severe malaria**: 4-hourly (2-hourly if unconscious or on quinine) and at any deterioration.
- **SAM**: every 2–4 hours on day 1; temperature with every check.
- **Neonates**: 1–3-hourly after a bolus until stable; before feeds for at least 24 h in IDM and SGA babies.
- **Conscious level** (AVPU/Blantyre) before and after dextrose.
- **Follow-up**: neurodevelopment after severe or recurrent neonatal hypoglycaemia; diabetes education review; ketotic hypoglycaemia counselling.

Prevention on the ward and at home

Group	Prevention
All sick children	Keep feeding (breastfeed, cup feeds, NG milk); avoid prolonged "nil by mouth"; maintenance fluids with dextrose if not feeding; check glucose in at-risk children
Severe malaria	4-hourly glucose (2-hourly if unconscious/quinine); artesunate rather than quinine; quinine in dextrose; early feeding
SAM	Feed every 2–3 h **including at night** (night feeds prevent most deaths), warmth, antibiotics
Newborns	Early and frequent breastfeeding, warmth, screen at-risk babies
Before surgery	Minimal fasting (clear fluids until 1–2 h before; breast milk until 4 h), dextrose-containing fluids
Diabetes	Education on symptoms, sugar always available, glucometer, glucagon if possible, regular meals, insulin adjustment for exercise and illness
Poisoning	Keep adult medicines (diabetes tablets) and alcohol out of reach

10. Complications

Disease-related

Complication	Notes
Death	Especially in severe malaria, SAM and neonates
Seizures, status epilepticus	Hypoglycaemia lowers the seizure threshold
Permanent brain injury	Occipital lobe damage (cortical visual impairment), epilepsy, developmental delay, learning problems

Treatment-related

Complication	Notes
Rebound hypoglycaemia	After a bolus without follow-up feeding or infusion
Thrombophlebitis / extravasation injury	Concentrated dextrose (>12.5% peripherally; 50% undiluted)
Hyperglycaemia and osmotic diuresis	Overtreatment, especially in neonates on high GIR
Fluid overload	Large-volume dextrose infusions in neonates or malnourished children
Vomiting	Glucagon

11. Prognosis

- Promptly treated hypoglycaemia recovers **within minutes**; the outcome then depends on the cause (severe malaria, sepsis, SAM).
- Hypoglycaemia in severe malaria is associated with a **several-fold increase in mortality**; dysglycaemia in critically ill Ugandan children is linked to higher mortality.
- Coma that persists after glucose correction means another cause (cerebral malaria, meningitis, post-ictal state, poisoning).
- Severe, prolonged or recurrent neonatal hypoglycaemia can cause **cortical blindness, epilepsy and developmental delay**.
- **Ketotic hypoglycaemia** usually resolves by school age.

12. Prevention

- **Every sick child**: keep feeding; avoid prolonged "nil by mouth"; dextrose-containing maintenance fluids if not feeding; check glucose in at-risk children (table in section 9).
- **Severe malaria**: artesunate rather than quinine; regular glucose checks; early feeding.
- **SAM**: 50 mL of 10% glucose or sugar water on admission; F-75 every 2–3 hours **day and night**; warmth; antibiotics.
- **Newborns**: breastfeed within 1 hour and every 2–3 hours; warmth (skin-to-skin, KMC); screen at-risk babies.
- **Diabetes**: education, sugar always available at home and school, glucometer, glucagon if possible, insulin adjustment for exercise and illness.
- **Ketotic hypoglycaemia**: avoid long fasts, bedtime carbohydrate snack, sugary drinks during illness.
- **Poisoning**: keep diabetes tablets and alcohol out of reach.

13. Special Populations

- **Neonates**: threshold <2.6 mmol/L (Ugandan units) or <2.2 (WHO Pocket Book); bolus 2 mL/kg; infusion 60 mL/kg/day on day 1; may be asymptomatic, so screen at-risk babies.
- **Severe acute malnutrition**: threshold <3.0 mmol/L; no glycogen or fat reserves and no warning symptoms; hypoglycaemia travels with hypothermia and infection; **no glucagon**; feed day and night.
- **Severe malaria and quinine**: quinine-induced hyperinsulinism; give in dextrose; check 2-hourly.
- **Type 1 diabetes**: threshold <3.9 mmol/L; rule of 15; glucagon useful (liver glycogen present).
- **Children on beta-blockers or with recurrent episodes**: may lack adrenergic warning signs.
- **Toddlers 18 months–5 years**: ketotic hypoglycaemia after an overnight fast or illness.
- **Before surgery**: minimal fasting and dextrose-containing fluids.

14. Key Points

High-yield points

- **Check glucose in every child who is unconscious, convulsing, malnourished, shocked or not feeding; if you cannot check, treat.**
- Thresholds: child **<2.5**, SAM **<3.0**, neonate **<2.6** (Uganda), diabetes **<3.9** mmol/L; severe malaria criterion **<2.2**.
- **10% dextrose 5 mL/kg** (neonate **2 mL/kg**) over 2–5 minutes.
- **Never 50% undiluted**: 1 part 50% + 4 parts water = 10%.
- **Recheck at 30 minutes** and **feed or infuse** to prevent rebound.
- No IV: NG sugar water 5 mL/kg, sublingual sugar every 10–20 min, IO.
- **Glucagon is useless without glycogen** (SAM, fasting, alcohol, liver disease).
- **No ketones → think insulin** (or a fat-burning defect).
- Quinine releases insulin; artesunate is first line.
- GIR >8–10 mg/kg/min in a neonate = hyperinsulinism.

Numbers & doses

Item	Value
Threshold, sick child	**<2.5 mmol/L** (45 mg/dL)
Threshold, SAM	**<3.0 mmol/L** (54 mg/dL)

Item	Value
Threshold, severe malaria criterion (UCG)	**<2.2 mmol/L** (40 mg/dL)
Threshold, neonate (Ugandan units)	**<2.6 mmol/L** (WHO Pocket Book 2.2)
Threshold, diabetes (ISPAD 2022)	Level 1 **<3.9**; level 2 **<3.0**; level 3 = needs help
Conversion	mmol/L × 18 = mg/dL
Bolus, child	**10% dextrose 5 mL/kg** (0.5 g/kg)
Bolus, neonate	**10% dextrose 2 mL/kg**
Making 10%	**1 part 50% + 4 parts water**; or remove 50 mL from 500 mL of D5 and add 50 mL of D50
Recheck	**30 min** (15 min in diabetes)
Oral treatment	**0.3 g/kg** fast sugar (10–20 g)
Sugar solution	**20 g (4 level teaspoons) in 200 mL** water
SAM admission	**50 mL** 10% glucose or sugar water, then F-75 every 30 min × 2 h, then 2-hourly
Glucagon	**0.5 mg (<25 kg); 1 mg (≥25 kg)** IM/SC
Neonatal maintenance day 1	**10% dextrose 60 mL/kg/day**
GIR	% × mL/kg/day ÷ 144; normal neonatal need **4–6 mg/kg/min**; **>8–10** = hyperinsulinism
Monitoring in severe malaria	Glucose **4-hourly** (2-hourly if unconscious or on quinine)
Persistent disorder targets (specialist)	>3.3 mmol/L first 48 h; >3.9 mmol/L after 48 h

Exam traps

- **"Glucose not measurable, so wait for the lab."** Wrong: **treat as hypoglycaemia** in any child with coma or convulsions.
- **"Give 50% dextrose 1 mL/kg as for adults."** Wrong: dilute to **10%** and give 5 mL/kg.
- **"Glucagon for a hypoglycaemic child with kwashiorkor."** Wrong: no glycogen; use IV/NG glucose.
- **"Glucose now normal; stop monitoring."** Wrong: rebound is common; **feed or infuse** and recheck.
- **"Hypoglycaemia in malaria is just from not eating."** Partly: also parasite use, impaired gluconeogenesis and **quinine-induced insulin release**.
- **"SAM child: use the 2.5 mmol/L threshold."** Wrong: **3.0 mmol/L**.
- **"Neonate: give 5 mL/kg like an older child."** Wrong: **2 mL/kg**, then an infusion.
- **"Hypoglycaemia with no ketones is ketotic hypoglycaemia."** Wrong: no ketones → **hyperinsulinism or FAOD**.

Questions seniors ask

Viva questions

1. **Define hypoglycaemia in a sick child.** Glucose <2.5 mmol/L (45 mg/dL) in a well-nourished child, <3.0 mmol/L in SAM (WHO); UCG 2023 notes symptoms usually occur below 3.0 mmol/L.
2. **Dose of dextrose?** 10% dextrose 5 mL/kg IV (2 mL/kg in neonates); recheck in 30 minutes.
3. **Why not 50% dextrose?** It is hyperosmolar (2,500 mOsm/L), sclerosing, can cause tissue necrosis if extravasated and causes rebound hypoglycaemia by a large insulin surge.
4. **How do you treat without IV access?** Sugar water or 10% dextrose by NG tube, sublingual sugar every 10–20 minutes, IO access, glucagon if insulin-induced.

5. **Name five causes of hypoglycaemia in Ugandan children.** Severe malaria (± quinine), sepsis, SAM, prolonged fasting/vomiting, neonatal (preterm/SGA/IDM/asphyxia); also alcohol and sulphonylurea poisoning, insulin in diabetes.
6. **Why does an infant of a diabetic mother become hypoglycaemic?** Fetal hyperinsulinism from chronic exposure to maternal hyperglycaemia continues after the cord is cut.
7. **What is the critical sample?** Blood taken at the time of hypoglycaemia (before dextrose) for glucose, insulin, C-peptide, cortisol, GH, BOHB, free fatty acids, lactate and ammonia, plus the first urine for ketones and organic acids.
8. **When is glucagon useless?** When liver glycogen is depleted: SAM, prolonged fasting, alcohol, liver failure, glycogen storage disease type 1.

On the ward round

1. **This child is convulsing. What is your first investigation?** A bedside glucose. If I can't measure it, I treat with 10% dextrose 5 mL/kg.
2. **How much 10% dextrose for a 9 kg child, and how will you make it?** 45 mL: 9 mL of 50% dextrose plus 36 mL of water for injection.
3. **What do you do after the bolus?** Recheck glucose in 30 minutes, then feed or start a dextrose-containing maintenance infusion, monitor 2–4 hourly, and treat the cause.
4. **What if the glucose stays low after two boluses?** Start a 10% dextrose infusion and calculate the GIR; consider sepsis, adrenal insufficiency (hydrocortisone) or hyperinsulinism; call a senior.
5. **Why is hypoglycaemia so common in SAM?** No glycogen or fat stores, impaired gluconeogenesis, frequent infection and hypothermia, and long gaps between feeds (especially at night).
6. **Which newborns do you screen?** Preterm, LBW/SGA, infants of diabetic mothers, asphyxiated, septic, hypothermic or poorly feeding babies: check before the second feed and before feeds for at least 24 hours.
7. **How would you counsel the mother of a child with ketotic hypoglycaemia?** Avoid long fasts, give a bedtime carbohydrate snack, give sugary drinks frequently when the child is ill or not eating, and come to hospital early if the child is drowsy or vomiting; it usually resolves by school age.

Mnemonics

- **"2–3–2.6–3.9"**: child 2.5, SAM 3.0, neonate 2.6, diabetic 3.9 mmol/L.
- **"5-2-30"**: 5 mL/kg for children, 2 mL/kg for neonates, recheck in 30 minutes.
- **"1 + 4 = 10"**: 1 part 50% dextrose + 4 parts water = 10%.
- **"Rule of 15"** (diabetes): 15 g fast sugar, recheck in 15 minutes.
- **Causes "HIGH SUGAR" is what they need**:
  - **H**ormones (cortisol, GH)
  - **I**nsulin excess (IDM, quinine, sulphonylurea, insulin)
  - **G**lycogen storage disease / liver failure
  - **H**unger (fasting, SAM)
  - **S**epsis
  - **U**nder-weight / preterm newborn
  - **G**ot malaria
  - **A**lcohol
  - **R**are metabolic defects (fatty-acid oxidation, galactosaemia)
- **Ketones**: "**No ketones, think insulin (or fat-burning defect)**."

OSCE tips

- In a "convulsing child" station, **say "I will check the blood glucose"** before you name the anticonvulsant; if no glucometer, "I will treat as hypoglycaemia".
- Show the calculation aloud: "**5 mL/kg × 12 kg = 60 mL of 10% dextrose**, made from 12 mL of 50% and 48 mL of water for injection."
- State the follow-up: **recheck in 30 minutes, feed or infuse, find the cause**.

Presenting this tutorial

> NOTE
> **Edith presents this tutorial** with **Ampaire Perez**, moderated by **Dr Mawanda** (Week 4, Tuesday, 08:00–09:30). Students prepare and present; the tutor moderates and probes. Marks reward being **present, active and presenting** (3–5 points for presenters), so preparation and delivery both count.

Plan the 90 minutes

Time	Segment	Who	What to do
0–5 min	**Opening case**	Edith	Read a 4-line case (below) and ask: "What is the first thing you would do?" Let 2–3 colleagues answer.
5–15 min	**Definitions and thresholds**	Edith	The threshold table (child, SAM, malaria, neonate, diabetes); mmol/L ↔ mg/dL; Whipple's triad
15–25 min	**Physiology and causes**	Ampaire	Fed → fasting table; supply/use/production/hormones/insulin; the ketone question
25–35 min	**Clinical features and who to screen**	Ampaire	Adrenergic versus neuroglycopenic; neonatal signs; "check glucose in every…" list
35–55 min	**Management**	Edith	Algorithm; dextrose 5 mL/kg and 2 mL/kg; making 10% from 50%; no-IV options; follow-up feeds/infusion; malaria, SAM, neonatal and diabetes specifics
55–60 min	**Prevention and complications**	Ampaire	The prevention table; brain injury
60–75 min	**Back to the case + quick quiz**	Both	Finish the opening case; 5 rapid MCQs from this page
75–90 min	**Tutor discussion**	Dr Mawanda	Answer questions; note what the tutor emphasises

The opening case (read it aloud)

"A 3-year-old boy, 13 kg, is brought to Mengo at 2 a.m. with 2 days of fever and vomiting. He had a convulsion at home and is now responding only to pain. Hands warm, pulse 140, SpO_2 96%. The glucometer has no strips. **What do you do in the next 5 minutes?**"

Answer to reveal at the end: recovery position, oxygen not needed (SpO_2 96%), **10% dextrose 65 mL IV** (treat because you cannot measure), mRDT, Hb, IV artesunate 39 mg, recheck when strips arrive, maintenance dextrose fluid, look for meningitis.

Suggested slides or flip-chart pages (12–14)

1. Title, names, date, tutor.
2. Opening case.
3. Why it matters (brain fuel, children's high use, Ugandan causes).

4. Thresholds table.
5. Physiology diagram (fed → fasting; counter-regulation).
6. Causes by mechanism (supply, use, production, hormones, insulin).
7. Ketones: present versus absent.
8. Clinical features (two-column table) and neonatal signs.
9. Who to check (the "check glucose in every…" list).
10. Management algorithm.
11. Doses and how to make 10% dextrose (show the arithmetic for a 10 kg child).
12. Special groups: malaria/quinine, SAM, neonates, diabetes.
13. Prevention.
14. Summary: five take-home messages.

> CLINICAL PEARL
> **Bring props**: a 50% dextrose ampoule (empty), a 10 mL and a 50 mL syringe, a glucometer and strip, a teaspoon and sugar. Demonstrating the 1:4 dilution and the sublingual sugar technique makes the tutorial memorable and shows practical skill.

Questions Dr Mawanda is likely to ask (prepare answers)

- **"What threshold do you use in a child with SAM, and why is it higher?"** <3.0 mmol/L; they have no glycogen or fat reserves and lack warning symptoms, so they deteriorate quickly.
- **"How would you give glucose if the IV line fails?"** NG 10% dextrose or sugar water 5 mL/kg; sublingual sugar every 10–20 min; IO line.
- **"Why do children on quinine become hypoglycaemic?"** Quinine stimulates insulin release.
- **"The child's glucose is normal after treatment but he is still unconscious. What next?"** Look for cerebral malaria, meningitis (LP when safe), post-ictal state, poisoning; keep checking glucose.
- **"How do you prevent rebound hypoglycaemia?"** Feed or start a dextrose infusion immediately after the bolus; recheck in 30 minutes and regularly.
- **"What is ketotic hypoglycaemia?"** Fasting hypoglycaemia with high ketones in a thin toddler (18 months–5 years); diagnosis of exclusion; prevented by frequent carbohydrate and sugary drinks in illness; outgrown by school age.
- **"What is the GIR and why does it matter?"** % dextrose × mL/kg/day ÷ 144; >8–10 mg/kg/min suggests hyperinsulinism.

Delivery tips

- Speak to the room, not the slides; **pause for questions** after each section.
- Use **real numbers** (weights and volumes) every time you mention a dose.
- Divide the content evenly with Ampaire and **rehearse the handover** between sections.
- End with **five take-home messages**: (1) check glucose in every sick child, treat if you can't; (2) 10% dextrose 5 mL/kg (2 mL/kg neonate); (3) never undiluted 50%; (4) recheck at 30 minutes and feed or infuse; (5) prevent: feed, warm, artesunate, night feeds in SAM.
- Prepare a **one-page handout** with the threshold table, the algorithm and the drug table.

15. References

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SELF-TEST

Flashcards

26 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Hypoglycaemia treatment threshold in a well-nourished sick child (WHO)?	**<2.5 mmol/L (45 mg/dL)**.
2	Hypoglycaemia threshold in a child with severe acute malnutrition?	**<3.0 mmol/L (54 mg/dL)** (WHO, UCG 2023).
3	Hypoglycaemia as a criterion of severe malaria (UCG 2023)?	Blood glucose **<2.2 mmol/L (40 mg/dL)** with parasitaemia.
4	Neonatal hypoglycaemia threshold used in Ugandan newborn units?	**<2.6 mmol/L** (47 mg/dL). The WHO Pocket Book treats at <2.2 mmol/L. Symptomatic babies are treated regardless.
5	ISPAD 2022 levels of hypoglycaemia in diabetes?	**Level 1**: <3.9 mmol/L (alert; treat with fast sugar). **Level 2**: <3.0 mmol/L (clinically significant). **Level 3**: severe event needing another person's help (coma, convulsion).
6	Convert mg/dL to mmol/L for glucose.	Divide by **18** (e.g. 45 mg/dL = 2.5 mmol/L; 54 mg/dL = 3.0 mmol/L).
7	IV dextrose bolus for hypoglycaemia in a child?	**10% dextrose 5 mL/kg** (0.5 g/kg) IV over 2–5 min (UCG 2023 allows 2–5 mL/kg); recheck in **30 min**; repeat if still low.
8	IV dextrose bolus for a hypoglycaemic neonate?	**10% dextrose 2 mL/kg** IV, then 10% dextrose infusion (60 mL/kg/day on day 1); recheck after 30 min.
9	How do you prepare 10% dextrose from 50% dextrose?	**1 part 50% + 4 parts water for injection** (e.g. 1 mL/kg + 4 mL/kg). UCG alternative: remove 50 mL from a 500 mL bottle of 5% dextrose and add 50 mL of 50%. Make fresh; don't store.
10	Why never give 50% dextrose undiluted to a child?	Hyperosmolar and **sclerosing** (phlebitis, tissue necrosis if extravasated), and the sudden glucose load causes an insulin surge and **rebound hypoglycaemia**.
11	Options when IV access fails in a hypoglycaemic child?	**NG tube**: 10% dextrose or sugar water 5 mL/kg, or milk; **sublingual sugar** (1 level teaspoon moistened, every 10–20 min); **intraosseous** line; **glucagon IM** if insulin-induced.
12	How do you make sugar solution for NG use (WHO)?	**4 level teaspoons (20 g) of sugar in a 200 mL cup of clean water**, about 10%.
13	Glucagon dose for severe hypoglycaemia in a child on insulin?	**0.5 mg IM/SC if <25 kg; 1 mg if ≥25 kg** (about 20–30 µg/kg). May repeat once after 15 min. Causes vomiting.
14	When is glucagon ineffective?	When the liver has **no glycogen**: SAM, prolonged fasting, alcohol intoxication, liver failure, glycogen storage disease type 1.
15	Why does quinine cause hypoglycaemia?	It **stimulates insulin secretion** from pancreatic beta cells (hyperinsulinaemic hypoglycaemia). Give quinine in 5–10% dextrose and monitor glucose 2-hourly; prefer artesunate.
16	Four mechanisms of hypoglycaemia in severe malaria?	**Reduced intake** (vomiting, fasting), **increased use** (fever, parasites, hypoxic tissue), **impaired gluconeogenesis** (liver dysfunction, lactic acidosis), **quinine-induced hyperinsulinism**.
17	SAM admission protocol to prevent hypoglycaemia?	**50 mL of 10% glucose or sugar water** orally/NG on admission, then **F-75 every 30 min for 2 h**, then **every 2 hours day and night**; keep warm; antibiotics. Unconscious: 10% glucose 5 mL/kg IV, then 50 mL by NG.

#	Question	Answer
18	Low glucose with high ketones vs low glucose without ketones?	**High ketones**: starvation, ketotic hypoglycaemia, cortisol/GH deficiency, glycogen storage disease. **No ketones**: **hyperinsulinism** (IDM, quinine, sulphonylurea, insulin) or a **fatty-acid oxidation defect**.
19	What is ketotic hypoglycaemia?	Fasting hypoglycaemia with high ketones in a thin toddler (**18 months–5 years**) after an overnight fast or illness; diagnosis of exclusion; prevented by frequent carbohydrates; usually outgrown by school age.
20	What is the 'critical sample'?	Blood taken **at the time of hypoglycaemia, before dextrose**: glucose, insulin, C-peptide, cortisol, GH, BOHB, free fatty acids, lactate, ammonia (± first urine for ketones/organic acids). Never delay treatment for it.
21	Glucose infusion rate (GIR) formula?	**GIR (mg/kg/min) = % dextrose × mL/kg/day ÷ 144**. Normal neonatal need ≈4–6 mg/kg/min; **>8–10** suggests hyperinsulinism.
22	Adrenergic vs neuroglycopenic features?	**Adrenergic**: sweating, tremor, pallor, tachycardia, anxiety, hunger. **Neuroglycopenic**: irritability, lethargy, confusion, abnormal behaviour, seizures, coma.
23	Signs of hypoglycaemia in a newborn?	Jitteriness, poor suck, floppiness, apnoea, cyanotic spells, hypothermia, weak or high-pitched cry, seizures, or **no signs at all** (so screen at-risk babies).
24	Which newborns should be screened for hypoglycaemia?	Preterm, LBW/SGA, **infants of diabetic mothers**, asphyxiated, septic, hypothermic or poorly feeding babies: check before the second feed, then before feeds for at least 24 h.
25	Oral treatment of hypoglycaemia in a conscious child (diabetes 'rule of 15')?	**0.3 g/kg** fast-acting sugar (about 10–15 g), recheck after **15 minutes**, repeat if still <3.9 mmol/L, then a complex-carbohydrate snack.
26	Hypoglycaemia + shock + low sodium + high potassium in an infant?	**Adrenal crisis** (e.g. salt-wasting CAH): IV 0.9% saline, 10% dextrose, and **hydrocortisone** (infant 25 mg; or 2–4 mg/kg) IV, then 6-hourly.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 2-year-old boy (12 kg) with severe malaria is convulsing at Mengo. Glucose is 1.7 mmol/L. An IV line is in place. Which is the MOST appropriate glucose treatment?

A 50% dextrose 12 mL IV push
B 10% dextrose 60 mL IV over 2–5 minutes, then recheck in 30 minutes
C 5% dextrose 500 mL over 4 hours
D Glucagon 1 mg IM
E Oral sugar water 100 mL

2. A 16-month-old girl (6.4 kg, MUAC 10.8 cm, bilateral pedal oedema ++) is admitted to the nutrition unit. She is alert, drinking, temperature 35.9 °C, glucose 2.8 mmol/L. What is the BEST immediate management?

A No treatment because glucose is above 2.5 mmol/L
B Glucagon 0.5 mg IM
C 10% dextrose 5 mL/kg IV and keep nil by mouth
D 50 mL of 10% glucose or sugar water orally, then F-75 every 30 minutes for 2 hours, keep warm and start antibiotics
E Give RUTF and review in the morning

3. At a HC III, a 4-year-old (15 kg) is unconscious with fever. There are no glucometer strips. Two attempts at IV access have failed and no intraosseous needle is available. What is the BEST way to give glucose now?

A Pass an NG tube and give 75 mL of 10% dextrose or sugar solution, and give sublingual sugar while preparing
B Pour sugar water into his mouth with a cup
C Wait for strips to confirm hypoglycaemia before giving anything
D Give 50% dextrose 15 mL rectally
E Give glucagon 0.5 mg IM as the only treatment

4. A 2.8 kg term baby of a mother with gestational diabetes is jittery at 2 hours of age and not sucking well. Glucose is 1.8 mmol/L. What is the MOST appropriate management?

A Breastfeed and recheck in 3 hours
B 10% dextrose 5 mL/kg IV over 2 minutes
C 50% dextrose 1 mL/kg IV
D Glucagon 1 mg IM
E 10% dextrose 2 mL/kg IV, then a 10% dextrose infusion at 60 mL/kg/day with continued feeds and glucose rechecks

5. A 6-year-old (19 kg) on IV quinine for severe malaria (artesunate out of stock) becomes sweaty and confused on day 2. Glucose 2.0 mmol/L. After 10% dextrose 95 mL she wakes up. Which is the BEST explanation and plan? Advanced

A Cerebral malaria relapse; give a second loading dose of quinine
B Hypoglycaemia from vomiting only; give oral sugar and continue quinine in normal saline
C Quinine-induced hyperinsulinaemic hypoglycaemia; give quinine in 5–10% dextrose, feed, check glucose 2-hourly and switch to artesunate when available
D Adrenal insufficiency; give hydrocortisone
E Hypoglycaemia from paracetamol; stop paracetamol

6. A 3-year-old (14 kg) was treated with 10% dextrose 70 mL for a glucose of 1.9 mmol/L. At 30 minutes glucose is 5.8 mmol/L but he is still drowsy and vomiting. What is the MOST important next step to prevent recurrence?

A No further action; the glucose is now normal
B Give another 70 mL bolus of 10% dextrose
C Start a maintenance infusion containing 5–10% dextrose and recheck glucose 1–2 hourly
D Give glucagon 0.5 mg IM
E Give oral sugar water by cup

7. A 2½-year-old thin boy has had two morning seizures over 6 months, each after a day of poor eating during a cold. During the latest episode: glucose 2.0 mmol/L, urine ketones +++, no hepatomegaly, normal genitalia and growth along the 3rd centile. What is the MOST likely diagnosis? Advanced

A Ketotic hypoglycaemia
B Congenital hyperinsulinism
C Glycogen storage disease type 1
D Medium-chain acyl-CoA dehydrogenase (MCAD) deficiency
E Sulphonylurea poisoning

8. A 10-year-old girl (28 kg) with type 1 diabetes collapses at school after playing football and missing lunch. She is unconscious, and the school nurse has no IV access. What should the nurse give?

A Oral sugar water poured into her mouth
B Insulin, because she may be in DKA
C Nothing until she reaches hospital
D Glucagon 1 mg IM, then oral sugar and a snack when she wakes
E Glucagon 0.25 mg IM

9. A 2.0 kg preterm neonate is on 10% dextrose at 90 mL/kg/day and continues to have glucose readings of 2.0–2.3 mmol/L. What is the glucose infusion rate, and what does it suggest? Advanced

A 2.5 mg/kg/min: the infusion is inadequate, increase to 150 mL/kg/day immediately
B 6.25 mg/kg/min: above normal production; increase GIR carefully and consider hyperinsulinism or sepsis if needs exceed 8–10 mg/kg/min
C 9 mg/kg/min: confirms hyperinsulinism
D 62.5 mg/kg/min: dangerously high; stop the infusion
E 6.25 mg/kg/min: normal, so the glucometer must be wrong; ignore it

10. A 3-year-old (14 kg) is brought at 07:00 drowsy and sweaty. Last night his grandmother, who has type 2 diabetes, found her tablet strip open. Glucose 1.5 mmol/L; he improves after 10% dextrose 70 mL. What is the MOST important next step? Advanced

A Discharge after a meal because he is now awake
B Induce vomiting with salt water
C Give insulin to stabilise the glucose
D Give activated charcoal and discharge after 2 hours
E Admit for at least 24 hours of observation with a dextrose infusion and frequent glucose checks (octreotide if recurrent)

11. Which statement about making 10% dextrose on a Ugandan ward is CORRECT?

- A Mix equal volumes of 50% dextrose and water to make 10%
- B 50% dextrose can be given undiluted if pushed slowly
- C 5% dextrose is equivalent to 10% dextrose for boluses
- D It can be prepared in advance and stored for the week
- E 1 part 50% dextrose plus 4 parts water for injection gives 10% dextrose

12. A 5-week-old boy presents with vomiting, poor feeding and lethargy. Weight 3.2 kg (birth weight 3.4 kg), cold peripheries, CRT 4 s. Glucose 2.1 mmol/L, Na 121 mmol/L, K 7.0 mmol/L. His scrotum looks pigmented. What is the MOST important treatment in addition to fluids and dextrose? Advanced

- A Calcium gluconate only
- B IV hydrocortisone
- C IV ceftriaxone only
- D Insulin and dextrose for the potassium
- E Oral rehydration solution

13. Which group of hospitalised children is it MOST important to screen routinely for hypoglycaemia on admission?

- A Children with severe malaria, SAM, sepsis, coma or convulsions, and at-risk newborns
- B All children with simple URTI
- C Children admitted for elective hernia repair who are eating normally
- D Well children attending the immunisation clinic
- E Children with mild eczema

14. A 7-year-old (22 kg) is admitted unconscious after drinking a sachet of waragi at a party. He is cold (35.2 °C) and his glucose is 2.2 mmol/L. Which is the MOST appropriate management?

- A Glucagon 1 mg IM
- B Induce vomiting and observe
- C IV thiamine and no glucose until thiamine is given
- D 10% dextrose 110 mL IV, then a dextrose infusion, warming, airway protection and glucose monitoring
- E Flumazenil IV

15. During your tutorial, Dr Mawanda asks: 'Which finding at the time of hypoglycaemia most strongly suggests hyperinsulinism?' Which answer is BEST? Advanced

- A High urine ketones
- B Massive hepatomegaly with lactic acidosis
- C Low or absent ketones with detectable insulin and a high glucose requirement
- D Low cortisol
- E Hypoglycaemia only after 24 hours of fasting

Answers & explanations

> **1. B** · 10% dextrose 60 mL IV over 2–5 minutes, then recheck in 30 minutes
>
> B is correct: **10% dextrose 5 mL/kg** (12 × 5 = 60 mL) is the WHO/UCG dose, followed by a recheck at 30 minutes and a dextrose infusion or feeds. A is wrong: 50% dextrose must not be given undiluted to a child (sclerosis, rebound). C is wrong: 5% dextrose as slow maintenance is too little, too slowly, to correct a convulsion-causing hypoglycaemia. D is wrong: glucagon is for insulin-induced hypoglycaemia when IV access is impossible; malaria depletes glycogen. E is wrong: a convulsing child cannot swallow safely.

2. D · 50 mL of 10% glucose or sugar water orally, then F-75 every 30 minutes for 2 hours, keep warm and start antibiotics

D is correct: in **SAM** the threshold is **3.0 mmol/L**; an alert child gets **50 mL of 10% glucose or sugar water** by mouth, then F-75 frequently, warming (hypothermia) and antibiotics (hypoglycaemia + hypothermia suggest sepsis). A is wrong: 2.5 mmol/L is the threshold for well-nourished children. B is wrong: glucagon does not work without glycogen stores. C is wrong: IV dextrose is for the unconscious child; keeping her nil by mouth delays the feeds that prevent recurrence. E is wrong: RUTF is for the rehabilitation phase, not a complicated SAM child in stabilisation.

3. A · Pass an NG tube and give 75 mL of 10% dextrose or sugar solution, and give sublingual sugar while preparing

A is correct: when IV access fails, give **10% dextrose or sugar solution 5 mL/kg by NG tube** (15 × 5 = 75 mL); **sublingual sugar** is first aid while you prepare (WHO Pocket Book). B is wrong: liquid poured into an unconscious child's mouth will be aspirated. C is wrong: if glucose cannot be measured in a comatose child, treat as hypoglycaemia. D is wrong: rectal dextrose is poorly absorbed and not recommended. E is wrong: glucagon is unreliable in febrile, fasting, possibly malarial children and is not the first choice when an NG route is possible.

4. E · 10% dextrose 2 mL/kg IV, then a 10% dextrose infusion at 60 mL/kg/day with continued feeds and glucose rechecks

E is correct: a **symptomatic** neonate with glucose 1.8 mmol/L needs **10% dextrose 2 mL/kg** followed by an infusion (60 mL/kg/day on day 1) and continued feeding; IDMs have hyperinsulinism and will drop again without an infusion. A is wrong: feeding alone is only for asymptomatic, mildly low babies who can suck. B is wrong: 5 mL/kg is the dose for older children; neonates get 2 mL/kg. C is wrong: never give 50% dextrose to a baby. D is wrong: glucagon is not the first-line neonatal treatment and 1 mg is an adult-sized dose.

5. C · Quinine-induced hyperinsulinaemic hypoglycaemia; give quinine in 5–10% dextrose, feed, check glucose 2-hourly and switch to artesunate when available

C is correct: quinine stimulates **insulin release**; the plan is quinine diluted in dextrose, early feeding, 2-hourly glucose and artesunate (UCG first line) as soon as possible. A is wrong: the confusion resolved with glucose, and a repeat loading dose risks quinine toxicity. B is wrong: quinine should be diluted in dextrose, not saline, in a child prone to hypoglycaemia. D is wrong: nothing suggests adrenal insufficiency (no shock, no electrolyte disturbance). E is wrong: therapeutic paracetamol does not cause hypoglycaemia.

6. C · Start a maintenance infusion containing 5–10% dextrose and recheck glucose 1–2 hourly

C is correct: a bolus lasts minutes; a child who cannot eat needs a **dextrose-containing maintenance infusion** (Holliday–Segar about 48 mL/h for 14 kg) and frequent glucose checks to prevent **rebound hypoglycaemia**. A is wrong: rebound is common after a bolus without follow-up. B is wrong: another bolus is not needed when glucose is normal. D is wrong: glucagon is not a maintenance treatment and worsens vomiting. E is wrong: a drowsy, vomiting child may aspirate.

7. A · Ketotic hypoglycaemia

A is correct: fasting hypoglycaemia with **high ketones** in a thin toddler during illness, with a normal examination, is classic **ketotic hypoglycaemia** (a diagnosis of exclusion). B is wrong: hyperinsulinism suppresses ketone production, so ketones would be low. C is wrong: GSD type 1 causes massive hepatomegaly, lactic acidosis and hypoglycaemia after short fasts. D is wrong: MCAD deficiency causes hypoketotic hypoglycaemia. E is wrong: sulphonylureas cause hyperinsulinaemic (non-ketotic) hypoglycaemia, and the pattern tracks illness rather than access to tablets.

8. D · Glucagon 1 mg IM, then oral sugar and a snack when she wakes

D is correct: severe (level 3) hypoglycaemia in an insulin-treated child with no IV access: **glucagon 1 mg IM** (≥25 kg), then oral glucose and complex carbohydrate when awake (ISPAD 2022). A is wrong: unconscious children aspirate oral fluids. B is wrong: the story (exercise, missed lunch) is typical of hypoglycaemia; insulin could be fatal. C is wrong: delay risks brain injury. E is wrong: 0.25 mg is too low; children ≥25 kg get 1 mg (0.5 mg if <25 kg).

9. B · 6.25 mg/kg/min: above normal production; increase GIR carefully and consider hyperinsulinism or sepsis if needs exceed 8–10 mg/kg/min

B is correct: GIR = % × mL/kg/day ÷ 144 = 10 × 90 ÷ 144 = **6.25 mg/kg/min**, already at the upper end of normal neonatal glucose production; persisting hypoglycaemia needs a higher GIR and, if >8–10 mg/kg/min is required, evaluation for **hyperinsulinism** (and sepsis). A is wrong: the arithmetic is wrong and a sudden jump to 150 mL/kg/day risks fluid overload. C is wrong: the calculation is wrong and 6.25 does not yet confirm hyperinsulinism. D is wrong: misplaced decimal. E is wrong: low readings in a sick preterm baby must be acted on, ideally confirmed with a lab glucose.

10. E · Admit for at least 24 hours of observation with a dextrose infusion and frequent glucose checks (octreotide if recurrent)

E is correct: suspected **sulphonylurea** ingestion causes prolonged, recurrent hyperinsulinaemic hypoglycaemia (up to 24 hours or more, longer for long-acting agents); admit, infuse dextrose, check glucose 1–2 hourly, and use octreotide for recurrence. A is wrong: hypoglycaemia will recur. B is wrong: salt emetics can be fatal and emesis is not recommended. C is wrong: insulin would worsen hypoglycaemia. D is wrong: charcoal may help only soon after ingestion; early discharge is unsafe.

11. E · 1 part 50% dextrose plus 4 parts water for injection gives 10% dextrose

E is correct: 1 mL of 50% (0.5 g) + 4 mL water = 5 mL containing 0.5 g = **10%**; UCG also describes replacing 50 mL of a 500 mL bottle of 5% dextrose with 50 mL of 50%. A is wrong: equal volumes give 25% dextrose. B is wrong: undiluted 50% damages veins and causes rebound. C is wrong: 5% dextrose delivers half the glucose per mL and is not used for boluses. D is wrong: UCG says use immediately, do not store (infection risk).

12. B · IV hydrocortisone

B is correct: hypoglycaemia, **hyponatraemia, hyperkalaemia**, shock and hyperpigmentation in a young infant = **salt-wasting adrenal crisis (CAH)**; give **IV hydrocortisone** (infant about 25 mg or 50–100 mg/m²) with saline and dextrose. A is wrong: calcium protects the heart briefly but does not treat the cause. C is wrong: sepsis should be covered too, but antibiotics alone will not correct cortisol deficiency. D is wrong: insulin in a hypoglycaemic infant is dangerous; potassium falls with steroids and saline. E is wrong: a shocked infant needs IV treatment.

13. A · Children with severe malaria, SAM, sepsis, coma or convulsions, and at-risk newborns

A is correct: these are the groups in whom hypoglycaemia is common, often silent, and lethal: **severe malaria, SAM, sepsis, coma/convulsions and at-risk newborns** (preterm, SGA, IDM, asphyxia, hypothermia). B is wrong: well-feeding children with URTIs are at low risk. C is wrong: well-fed elective patients need only sensible fasting rules. D is wrong: routine screening of well children is not indicated. E is wrong: eczema does not cause hypoglycaemia.

14. D · 10% dextrose 110 mL IV, then a dextrose infusion, warming, airway protection and glucose monitoring

D is correct: alcohol blocks gluconeogenesis, so children develop **hypoglycaemia and hypothermia**; treat with **10% dextrose 5 mL/kg** (110 mL), a dextrose infusion, warming, recovery position and monitoring. A is wrong: glucagon is ineffective when alcohol blocks hepatic glucose output and glycogen is low. B is wrong: inducing vomiting in an unconscious child causes aspiration. C is wrong: thiamine-before-glucose applies to malnourished adult alcoholics, and must never delay glucose in a hypoglycaemic child. E is wrong: flumazenil reverses benzodiazepines, not alcohol.

15. C · Low or absent ketones with detectable insulin and a high glucose requirement

C is correct: insulin suppresses lipolysis and ketogenesis, so **hyperinsulinism** gives hypoglycaemia with **low ketones**, **detectable insulin** when glucose is low, and a **high glucose infusion requirement** (>8–10 mg/kg/min). A is wrong: high ketones point to starvation/ketotic hypoglycaemia or hormone deficiency. B is wrong: that describes glycogen storage disease type 1. D is wrong: low cortisol suggests adrenal insufficiency. E is wrong: hypoglycaemia only after a long fast suggests a fasting-adaptation problem; hyperinsulinism causes hypoglycaemia even after short fasts or soon after feeds.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: The tutorial opening case: a convulsing child at 2 a.m.

A 3-year-old boy is brought to Mengo at 02:00 with 2 days of fever and vomiting. He had a 15-minute convulsion at home and now responds only to pain. Hands warm, CRT 2 s, pulse 140/min, RR 36, SpO_2 96% in air. No wasting or oedema. Estimated weight 13 kg. The glucometer has no strips.

1. Triage him and state your first actions.

> **Emergency** (coma: AVPU P). Airway: recovery position, suction if needed. Breathing: SpO_2 96%, no oxygen required. Circulation: not shocked (no bolus). IV access and blood for **mRDT/smear, Hb, cross-match, and glucose if a lab can do it**.

2. There are no strips. What do you do about glucose?

> **Treat as hypoglycaemia**: 10% dextrose **5 mL/kg = 65 mL** IV over 2–5 min, made from **13 mL of 50% dextrose + 52 mL water for injection**. Document the time; recheck when strips or the lab result arrive (lab glucose comes back 1.6 mmol/L).

3. Thirty minutes later he is responding to voice. What next?

> Prevent rebound: **maintenance fluid containing dextrose** (e.g. 10% dextrose in 0.45% saline; Holliday–Segar 1,150 mL/day ≈ 48 mL/h) until he can feed; start feeds as soon as safe. Glucose **4-hourly** (2-hourly while drowsy). mRDT positive → **IV artesunate 3 mg/kg = 39 mg** at 0, 12, 24 h, then daily until oral ACT.

4. Why was this child hypoglycaemic?

> **Severe malaria**: reduced intake (2 days of vomiting), increased use (fever, parasites), impaired hepatic gluconeogenesis, plus the convulsion. If he had received **quinine**, hyperinsulinism would add to it. Also consider sepsis/meningitis.

5. Six hours later he is fully alert, eating porridge, glucose 6.1 mmol/L. How do you continue and when do you stop monitoring?

> Reduce and stop IV fluids as oral intake is reliable; continue glucose checks 4–6 hourly until he has eaten normally for 24 hours; complete artesunate then oral ACT; check Hb; HIV status; bed nets and early treatment advice; immunisation check.

6. Give the five take-home messages you would close the tutorial with.

> (1) Check glucose in every sick child; **treat if you can't measure**. (2) **10% dextrose 5 mL/kg** (2 mL/kg neonate). (3) **Never undiluted 50%**; 1 part + 4 parts water. (4) **Recheck at 30 min** and feed or infuse. (5) **Prevent**: feed, warm, artesunate, night feeds in SAM, screen at-risk newborns.

Take-away: In a comatose febrile child, glucose comes before the diagnosis; if you cannot measure it, treat it, then prevent rebound and treat the cause.

Case 2: Hypoglycaemia and hypothermia in kwashiorkor

A 20-month-old girl is admitted to the nutrition unit at 21:00. Weight 7.0 kg, length 76 cm, MUAC 11.2 cm, bilateral pitting oedema up to the knees, flaky-paint skin. She is lethargic but rousable, temperature 35.3 °C (axillary), glucose 2.4 mmol/L. Mother is HIV-positive on ART; the child's status is unknown.

1. What are her immediate problems?

> **Complicated SAM** (oedematous malnutrition, grade ++), **hypoglycaemia** (<3.0 mmol/L threshold in SAM), **hypothermia** (<35.5 °C), likely **sepsis**, HIV exposure.

2. How do you treat the hypoglycaemia?

> She is rousable and can swallow: **50 mL of 10% glucose or sugar water** (1 rounded teaspoon of sugar in 3 tablespoons of water) orally or by NG, then **F-75 every 30 minutes for 2 hours**, then **2-hourly day and night**. If she were unconscious: **10% glucose 5 mL/kg = 35 mL IV**, then 50 mL by NG. **No glucagon**.

3. How do you treat the hypothermia?

> Skin-to-skin (kangaroo) with the mother, cover including the head, warm room, warm blankets; avoid hot-water bottles directly on the skin; feed; recheck temperature every 30 min–2 hours; keep away from windows and draughts at night.

4. What other treatment starts tonight?

> **Antibiotics** (e.g. IV ampicillin 50 mg/kg 6-hourly + gentamicin 7.5 mg/kg daily for complicated SAM), no IV fluids unless shocked, careful hydration assessment (ReSoMal only if dehydrated), micronutrients (no iron yet), **HIV test** (at 20 months a rapid antibody test, confirmed per the national algorithm; DNA PCR is used under 18 months), skin care.

5. Why do most SAM deaths happen at night, and how do you prevent them?

> Long gaps between feeds overnight, cold rooms and fewer staff lead to **hypoglycaemia and hypothermia**. Prevention: **2–3-hourly feeds including at night**, warmth, glucose and temperature checks, and staff/caregiver education.

> **Take-away:** In SAM, treat glucose below 3.0 mmol/L, never use glucagon, and remember that hypoglycaemia, hypothermia and infection come together; night feeds save lives.

Case 3: A jittery baby of a diabetic mother

A 4.3 kg baby boy is born at 39 weeks by caesarean section to a mother with gestational diabetes on insulin. At 90 minutes of age he is jittery and has not latched. Temperature 36.6 °C, RR 52, no distress. Heel-prick glucose 1.9 mmol/L.

1. Why is this baby at risk of hypoglycaemia?

> **Infant of a diabetic mother**: fetal **hyperinsulinism** from exposure to maternal hyperglycaemia (Pedersen hypothesis) persists after the cord is cut, so glucose falls sharply in the first hours; macrosomia confirms poor maternal control.

2. What do you do now?

> He is **symptomatic** and <2.6 mmol/L: **10% dextrose 2 mL/kg = 8.6 mL IV** over 5 min, then **10% dextrose infusion 60 mL/kg/day** (≈11 mL/h), keep warm, continue/assist breastfeeding (EBM by cup or NG), recheck glucose in 30 min.

3. Glucose at 30 minutes is 2.4 mmol/L and later 2.2 mmol/L despite the infusion. What next?

> Calculate GIR: 10 × 60 ÷ 144 = **4.2 mg/kg/min**. Increase GIR stepwise (increase volume to 80–90 mL/kg/day or concentration to 12.5% peripherally), repeat a 2 mL/kg bolus if symptomatic, and inform the senior. A GIR >8–10 mg/kg/min would suggest persistent hyperinsulinism needing specialist care.

4. What other problems do you look for in this baby?

> **Hypocalcaemia**, **polycythaemia** (and jaundice), **respiratory distress** (delayed lung maturity, CS), **birth trauma** (shoulder dystocia, brachial plexus injury), congenital heart disease (septal hypertrophy), and **hypomagnesaemia**.

5. When can you stop monitoring?

> When he has had **three consecutive pre-feed glucose readings ≥2.6 mmol/L** on full enteral feeds after the infusion is weaned (usually within 24–48 h). Counsel the mother about postpartum glucose testing (OGTT at 6–12 weeks) for her own type 2 diabetes risk.

Take-away: Infants of diabetic mothers have transient hyperinsulinism: treat symptomatic hypoglycaemia with 2 mL/kg of 10% dextrose and an infusion, calculate the GIR, and keep feeding.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.